

ALBERT HO

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EDUCATION

Bachelor of Science, Computer Science — University of California, San Diego
Awards: Summer URS Philip and Elizabeth Hiestand Scholarship

Expected: June 2026
GPA: 3.8

EXPERIENCES

Software Engineer Research Intern | UCSD Health — Hojun Li Lab

October 2024 - Present

- Engineered a scalable Python-based bioinformatics pipeline for parsing BAM alignment files from Oxford Nanopore's Dorado platform, enabling efficient detection and quantification of A-to-I RNA editing events
- Developed a computationally optimized algorithm to calculate per-base modification frequencies, leveraging ONT's statistical models and vectorized Pandas/Numpy operations to minimize runtime and memory overhead
- Implemented automated genomic feature annotation using interval tree data structures, producing reproducible, research-ready datasets that support downstream analyses
- Engineered a reproducible, scalable workflow for more than 57,000 single-cell profiles using R, Seurat, and Python/Pandas, improving pipeline maintainability and computational efficiency

Data Science Intern | Lawrence Livermore National Laboratory

July 2024 - August 2024

- Developed a production-ready machine learning pipeline for ventricular arrhythmia classification, benchmarking Logistic Regression, Random Forest, and deep learning models (CNNs and RNNs)
- Achieved 99% recall by designing and hyperparameter-tuning a sequence-to-sequence convolutional neural network, significantly improving diagnostic sensitivity
- Processed and labeled large datasets (over 16,000 samples) using Pandas dataframes for training and testing

PROJECTS

Hall Pass | TypeScript, React Native, Firebase, AWS S3, Expo

January 2026 - Present

- Designed a full-stack social media platform for universities, managing the project lifecycle from UI design to deployment
- Implemented backend using Firebase for real-time messaging and AWS S3 for image storage, implementing private course-specific discussion spaces and verification
- Utilized AI-augmented development workflows to assist feature prototyping and optimize TypeScript codebases, while maintaining system logic and security protocols

GuitarMax | Swift, AVFoundation, SwiftUI

December 2025 - Present

- Developed an iOS application for real-time guitar education, using Apple's Vision framework to provide instant feedback on finger placement
- Built an interactive chord library using the Combine framework for reactive UI updates, allowing for low-latency visual overlays of fret and string guidelines on a live camera feed
- Leveraged AI assistance for rapid debugging and boilerplate generation, while fine-tuning a 12-tone coordinate mapping

Stock-em | Python, Flask, PyTorch, yfinance, HuggingFace, HTML, JS

October 2025 - Present

- Developed a LSTM, RNN hybrid model to predict futures using historic stock prices and news sentiment scoring
- Incorporated a backtesting model to benchmark predicted values on existing data; displayed the model's buy, sell, and short calls, as well as outcomes such as profits, losses, SHARPE ratio, and return percentages
- Implemented a full stack web application capable of allowing users test the pipeline on different tickers and view returns

TECHNICAL SKILLS

Languages — Python, R, C/C++, HTML, CSS, JavaScript, Java

Frameworks — Agile, React, Node.js, WordPress, Tailwind CSS

Tools & Libraries — Git, Github, VS Code, Firebase, AWS, PyCharm, XCode, Unreal Engine, scikit-learn, tensorflow, PyTorch, pandas, NumPy, Matplotlib, D3.js

Certifications — CodePath Certificate in Intro to Android Mobile Development

Relevant Coursework — Data Structures and Algorithms, Operating Systems, AI: Probabilistic Models, Software Engineering, Engineering Probability and Statistics, Intro to Cryptography, Theory of Computation